No of Pages : 5 **Course Code : 23MX14**

Roll No:
(To be filled in by the candidate)

PSG COLLEGE OF TECHNOLOGY, COIMBATORE 641 004
SEMESTER EXAMINATIONS, DECEMBER 2024
MCA Semester: 1
23MX14 DATABASE SYSTEMS

Time : 3 Hours **Maximum Marks:100**

INSTRUCTIONS:

1. Answer **ALL** questions. Each question carries 25 Marks.
2. Subdivisions (a) carries 3 marks each, subdivision (b) carries 10 marks each and subdivision (c) carries 12 marks each.
3. Course Outcome : Table

Qn.1	CO1	Qn.2	CO2	Qn.3	CO3	Qn.4	CO4

COs	At the end of learning this course, students will be able to
CO1	Exemplify the architecture and components of a database system
CO2	Design, manipulate and manage database for real world applications.
CO3	Analyse the data models, database transactions and security issues for real world problems
CO4	Demonstrate their understanding of key notions of relational data model and extend normalization for the development of application for the benefit of the society.

		Question	(BTL)
1.	a)	What is data redundancy? How does it impact a database?	(L2)
	b) i)	Exemplify the Three schema architecture with neat diagram. How do we achieve the logical data independence and physical data independence? (5)	(L3)
	ii)	Consider you want to build a video site similar to YouTube. The actual video data, the metadata about the video, such as title, the user who uploaded it must be stored. Is it better to use a file processing system or a database for storage? Justify your answer. (5)	
	c) i)	Design an Entity Relationship Diagram that models the following objects and relationships in the worldof football: teams, players, games, managers and contracts.	(L5)

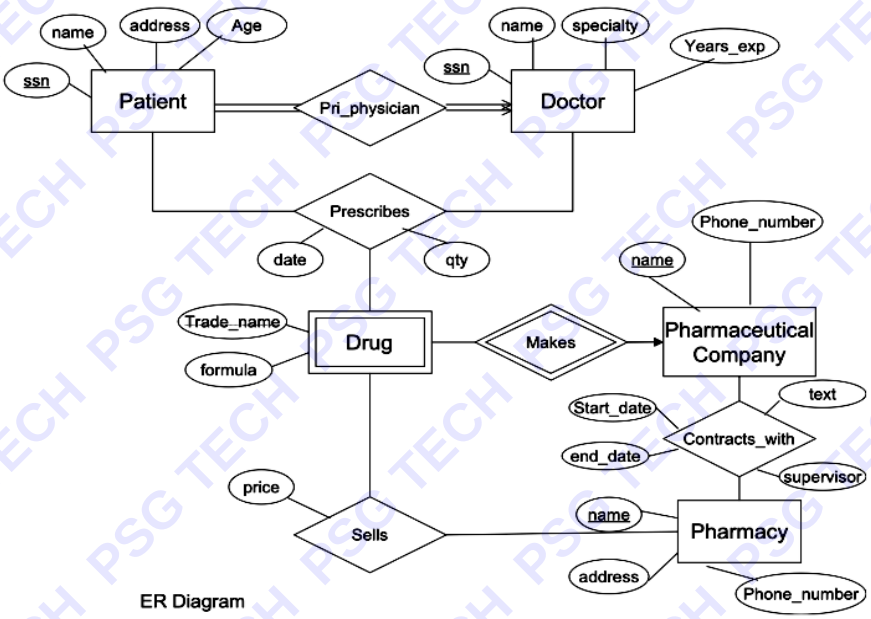
- Each team has a unique team name, and a city it plays in.
- Each person has a unique ssn and a name. Additionally, for players their weight, height, position and birth dates are of importance.
- Players have a contract with at most one team and receive a salary for their services.
- Each team can have more than one manager; managers can work for more than oneteam and receive a salary for each of their employments.
- Players cannot be managers.
- A game involves a home-team and visiting-team; additionally, the day of the game, andthe score of the game are of importance; teams play each other several times in a season (not on the same day).
- For each game played we like to know which players participated in the game and how many minutes they played. Indicate the cardinalities for

each relationship type; assign roles (role names) to eachrelationship if there are ambiguities. State clearly any assumptions made.

(OR)

ii) Consider the ER model given in Figure. This model represents the operations of a pharmacy chain. Answer the following questions regarding this model.

- Can a pharmaceutical company have multiple phone numbers? If not, what do you need to do to allow this?
- If we delete from the database the pharmaceutical company that manufactures a drug, what happens to the drugs that the company manufactures? Justify your argument.
- What if we delete the pharmacy that sells the drug?. Do we have to delete the drug too? Why or why not?



ER Diagram

- Convert ER diagram to relational tables using any relational model product by stating the steps and rules.

2. a) Compare the superkey, candidate key and primary key. (L2)

b) i) State the properties of relation and any five Codd's rules of relational data model. (5) (L4)

ii) Exemplify the significance of enforcing integrity constraints in a relational data model with suitable examples.

Consider a relation R: Initially, the relation is created using primary key and salary must <=Rs.40000 and has three records;

Name	Salary (in Rs)
Abi	10000
James	20000
Sita	30000

Write table creation statement and execute the following sequence of modificationstatements. Some of them may be rejected due to the

constraints in the relation.

- INSERT INTO R VALUES ('Fred',12000);
- UPDATE R SET salary = 50000 WHERE name = 'Abi';
- INSERT INTO R VALUES ('Sita', 13000);
- DELETE FROM R WHERE name = 'Abi';

At the end of these statements, analyse the result of the query to find the sum of the salaries over all the tuples in R. (5)

c) Exemplify the significance of relational algebraic expression used in query optimization. (L5)

Consider the following schemas and construct relational algebraic expression for the below questions.

Employee (empnum, empname, age, monthlysalary, deptnum)
Department (deptnum, deptname, deptbudget, hodname)
Project (projnum, projname, deptnum, projbudget)

i) List the projname, deptname, projbudget, deptbudget for projbudget value greater than 100000.

ii) List the empname, monthlysalary, deptname for deptbudget value greaterthan100000.

iii) For each department list the highest paid employee name and salary.

iv) Find the name of the department(s) where there is no employee.

v) Find the employees with their relevant department using natural join.

3. a) Consider the SQL query whose answer is shown in the table. (L3)

sid	Name	login	age	gpa
100	Madan	madan@music	11	1.8
101	Guru	guru@music	12	2.0

i) Modify this query so that only the login and gpa columns are included in the answer.

ii) If the clause WHERE S.gpa>= 2 is added to the original query, what is the set of tuples in the answer?

b) i) State the purpose of using views instead of accessing data directly from the database? Is it more efficient to use views or access tables directly? (L3)

Given the two relations:

Book(ISBN PK, title, pagecount, pubName FK ref(publisher))

Publishers(pubName PK, city, state)

Write the SQL statement for a virtual view named Bookcities. It should have the title and publishing city of every book. (5)

ii) Paraphrase the use of sequence, synonyms and indexing with examples. (5)

c) Illustrate the various types of joins with suitable example. (L5)

Consider you are designing a database for the Humane Society. The following is the set of relations with each relation's attribute is given following the attribute(e.g., ID: integer):

Animals(ID: integer, Name: string, PrevOwner: string, DateAdmitted:date, Type: string)

Adopter(SIN: integer, Name: string, Address: string, OtherAnimals:integer)

Adoption(AnimalID: integer, SIN: integer, AdoptDate: date, chipNo: integer)

Design the table using relevant integrity constraints and formulate the following queries in SQL;

i) Retrieve the total number of dogs that were brought to the Humane Society on 18 April 2000.

ii) List the name of the adopter who has adopted every type of animal.

iii) For each animal type, list the animal type and total number of adoptions on 14 June 2024.

iv) List the types of animals who have not had any adoptions.

4. a) Define partial and transitive functional dependency (FDs) with examples? (L2)

b) i) Consider a relation R(A, B, C, D, E) with FD's AB → C, CD → E, C → A, C → D, D → B. (L4)

The possible candidate keys for R are AB, AD, C

- List all the functional dependencies that violate 3NF. If any, then decompose R accordingly.
- List all the FD's that violate BCNF. If any, then decompose R accordingly. (5)

ii) State the security and integrity threats in the database system. Compare and contrast the discretionary access control and mandatory access control. (5)

c) i) Exemplify the need of normalizing the tables. The role of the government pharmacy is to import medicine into the country and distribute it to major hospitals around the country. Table shows the records that are kept by the government pharmacy when it distributes the medicine to the major hospitals around state. You are required to design a relational database to store the data shown in the table without any modification anomalies. (L5)

Attribute Name	Sample Value	Sample Value	Sample Value	Sample Value	Sample Value	Sample Value	Sample Value
Hospital_Num	H101	H101	H101	H101	H102	H102	H102
Hospital_Name	Suva Private	Suva Private	Suva Private	Suva Private	CWM	CWM	CWM
Med_Num	M1006	M1006	M1007	M1008	M1005	M1009	M1010
Med_Name	Throat Coat	Throat Coat	SinuFix	Panadol	Tylenol	Frova	Axert
MedType_ID	LQD	LQD	CAP	CAP	TBL	TBL	TBL
MedType_Desc	Fuild	Fuild	Capsules	Capsules	Tablets	Tablets	Tablets
Dispatch_Date	40182	40240	40190	40262	40212	40254	40240
Dispatch_Qty	10	10	50	50	100	10	20
Unit_ID	MU3	MU3	MU2	MU2	MU1	MU1	MU1
Unit_Desc	Bottles	Bottles	Boxes	Boxes	Cards	Cards	Cards

- Draw the dependency diagram showing all types of dependencies that are functional, partial, and transitive.
- Analyse the modification anomalies related to this table design.
- Convert un-normalized table structure to 3NF, and draw the new dependency diagram.

(OR)

Illustrate the ACID properties of transaction processing with example.

ii) Consider the following database schema: STUDENT(name, sid, gpa, level, dept) Suppose the following two transactions are executed concurrently:

T1	T2
begin trans update STUDENT set gpa = 4.0 where dept = 'CS' commit trans	begin trans insert into STUDENT values ('Ramya', 101, 3.9, 4, 'CS') ; insert into STUDENT values ('Suma', 102, 3.9, 3, 'CS') ; commit trans

Assume Ramya and Suma were not in the STUDENT table before the start of T1 or T2. Suppose read locks are released immediately after the read is done and write locks are held until end of transaction. Can it ever happen that after both T1 and T2 have committed, Ramya and Suma have different gpa values? State your reasoning in support of your conclusion. If your answer depends on locking granularity, access methods or indexing, then analyze the possibilities.

CO : COURSE OUTCOME; BTL : BLOOM'S TAXONOMY LEVEL

/END/

FD/RL